Experiment # 6

To Determine the Value of an Unknown Resistance and to Verify the Laws of Series & Parallel Resistance by Means of a Post Office Box

A. Determination of the value of an unknown resistance

Theory

If P and Q are the known resistances in the ratio arms and R that in the third arm (See Figure 1), the unknown resistance S in the fourth arm is obtained, when there is no deflection of the galvanometer, from the relation

$$\frac{P}{Q} = \frac{R}{S}$$

$$\text{Or, } S = \frac{RQ}{P}$$

Apparatus

P.O. Box-
Unknown Resistance(s)
Zero-Centre Galvanometer
Cell
Commutator
Connecting Wires

Description of the Apparatus

See the Post Office Box supplementary handout.

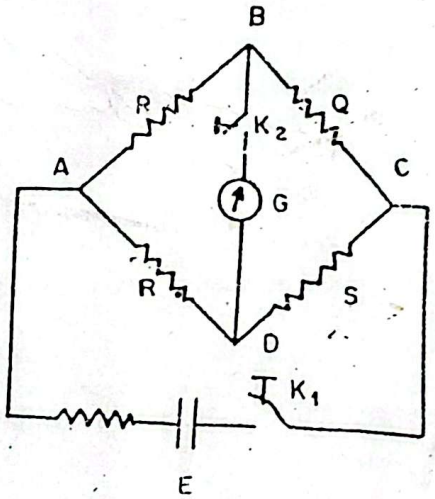

FIGURE 1

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Connect the terminals of the galvanometer between D and K_2 of the P.O. Box (Figure 1) K_2 being internally connected to the point B. Connect the poles of the cell E through a rheostat Rh to the point K_1 and C, K_1 being internally connected to A. Connect the terminals of unknown resistance S to points C and D.

2. Take out the resistances 10 and 10 from the ratio arms BA and BC. *See that all the other plugs in the box are tight.* This means *zero resistance* in the third arm. Put the maximum resistance in the rheostat. Press the battery key K_1 and then press the galvanometer key K_2. Observe the direction of the deflection of the galvanometer. Next take out the infinity plug from the third arm and press the keys as done before. If opposite deflection is obtained then the connection is correct. If not then check the connections again.
3. Then go on gradually reducing the resistance in the third arm until a resistance say R_1, is found for which there is no deflection in the galvanometer when the circuit is closed. Then the unknown resistance S is given by
   $S = 10/10\ R_1 = R_1$ (say 5 ohms).
4. If instead of a null point, there is a deflection in one direction with R_1 and an opposite deflection with $(R_1 + 1)$ in the third arm, the unknown resistance is partly integral and partly fractional i.e., *it lies between 5 and 6 ohms.*
5. Now take out the resistance of 100 ohms in the arms P (BA) keeping 10 ohms in the arm Q (BC) so that the ratio is now Q/P = 10/100 = 1/10. Hence the null point should occur when the resistance in the third arm is of some value between 10 R_1 and 10 $(R_1 + 1)$ i.e., between 50 and 60 (if $R_1 = 5$). Observe the opposite deflection and as before *narrow down the range* to obtain the null point at $R_2 = 53$ (say). Then S = 53/10 = 5.3 ohms. In that case, *the resistance is found correct to one decimal place.*
6. If the null point cannot be obtained at this stage also i.e., if the opposite deflections are observed for R_2 and $R_2 + 1$ (viz for 53 and 54) in the third arm, *it lies between 5.3 ohms and 5.4 ohms.* Repeat the observations with 1000 ohms in P arm and 10 ohm in the Q arm. The resistance in the third arm should be between 530 and 540 for which opposite deflections will be obtained. Narrow down the range to obtain a bull point at $R_3 = 535$ (say). Then S = R/100 = 5.35 ohms (say). *The resistance is now correct to two decimal places.*
7. If even at this stage there are opposite deflections for a change of resistance of 1 ohm in the third arm, the unknown resistance can be determined to the third decimal place by proportional parts. But it is futile to expect such accuracy from the P.O. Box. However, if it is desired to go further, proceed as follows: Count the number of divisions for which the galvanometer is deflected when R_3 is put in the third arm. Suppose it is d_1 divisions to the left. If now for $(R_3 + 1)$ in the third arm, the deflection is d_2 to the right, then for a change of 1 ohm in the third arm, the pointer moves through $d_1 + d_2$ divisions. Hence to bring the pointer to zero of the scale (i.e., for no deflection), resistance $R_3 + \frac{d_1}{d_1 + d_2}$ is to be inserted in the third arm. Hence the value of the unknown resistance S is given by

$$S = \frac{1}{100} R_3 + \frac{d_1}{d_1 + d_2}$$

8. While taking the final reading with the ratio 1000:10, reverse the current and take the mean value of S.

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.umerical values in the example below are not actual readings, they are for .istration only. (Please use the table in the next page to record your readings). (a) and (b) are direct and reverse readings.

Resistance in ohms			Direction of Deflection	Third arm resistance is
Arm Q	Arm P	Arm R		
10	10	0	Left	Too small
			Right	Too Large
		100	"	"
		50	"	"
		20	"	"
		10	"	Large
		7	"	"
		6	"	"
		5.	Left	Small
				The unknown resistance lies between 5 & 6 ohms
10	100	60	Right	Large
		50	Left	Small
		55	Right	Large
		54	Right	"
		53	left	Small
				The unknown resistance lies between 5.3 ohms & 5.4 ohms
10	1000	539	Right	Large
		531	Left	Small
		533	"	"
		534	"	"
		535	7 divisions to the left	
		a{	} 14 divisions to the right	S=1/100(535+7/17) = 5.354 ohms
		536		
		536	8 divisions to the left	
		b{	}	S=1/100(535+6/14) = 5.354 ohms
		535	6 divisions to the right	

Mean S = 5.354 ohms

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To Verify the Laws of Series and Parallel Resistances

Theory

Resistances are said to be connected in series when they are connected with the end of one joined to the beginning of the next and so on and as shown in Figure 2 (a) The equivalent resistance to a number of resistances connected in series is equal to the sum of the individual resistances, i.e.,

$$R_s = r_1 + r_2 + r_3 + \ldots \quad (1)$$

When resistances are arranged with their respective ends connected to common terminals, they are said to be connected in parallel as shown in figure 2 (b). The reciprocal of equivalent resistance to a number of resistances connected in parallel is equal to the sum of the reciprocals of the individual resistances, i.e.,

$$\frac{1}{R_p} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \ldots \quad (2)$$

Measuring r_1, r_2, r_3 etc separately and the equivalent resistance R by connecting them in series and in parallel, the relation (1) and (2) may be verified.

Procedure

1. Measure the resistances r_1, r_2, r_3 etc separately by means of a P.O. Box as in part A of this experiment.
2. Join the resistances r_1, r_2, r_3 in series as in Figure 2 (a) and determine the equivalent resistance of the series combination by means of the P.O. BOX. Show that relation (1) holds good.
3. Connect the resistances in parallel as in Figure 2 (b) and determine the equivalent resistance of the parallel combination as before. Show that relation (2) holds good.

Results

Record data for r_1, r_2, r_3 etc and for the combination in the same tabular form as in part A of this experiment. From the observed and calculated values of the equivalent resistances thus obtained, show that they are equal within the limits of the experimental error. This verifies relation (1) and (2).

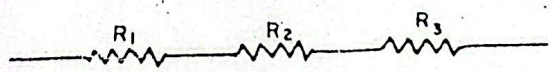

FIGURE 2(a)

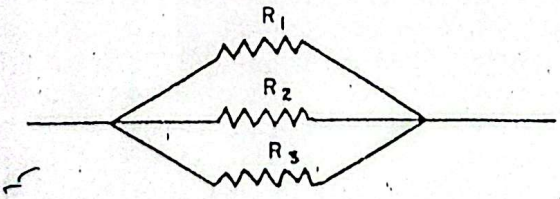

FIGURE 2(b)

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.O. Box) is another form of Wheatsone's bridge. It consists of a number of fixed resistancess in series much the way as resistance box of larger type (Fig. 7.29). A plan of this box is

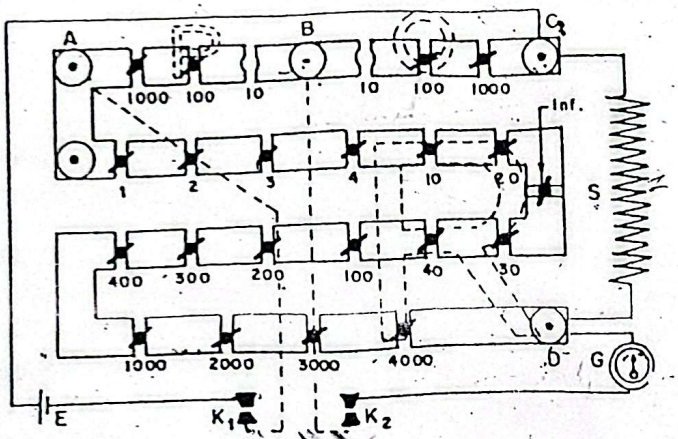

FIGURE 7.29

seen in Fig. 7.29. The group of resistance may be divided into three parts P, Q, and R called the arms of the P. O. Box. Each of the two arms P and Q contains a series of resistances 10, 100 and 1000 ohms interconnected by thick metal studs. These arms are called *ratio arms*. At each end of these arms there is a binding screw (*i.e.*, at A and C). The third arm R contains a number of resistance coils usually ranging from 1 to 4000 ohms. These resistances are arranged in such a way that any integral value of resistance from 1 to 11,110 ohms may be put in this arm. At the place marked *inf.* there is a plug called the *infinity plug*. The free end of this arm at D is provided with another binding screw. At the bottom, there are two tapping keys K_1 and K_2 provided with binding screws. The keys on being depressed make metallic contacts with the points A and B. *Students should note carefully that in some apparatus the key K_1 makes contact with the point C, not the point A.*

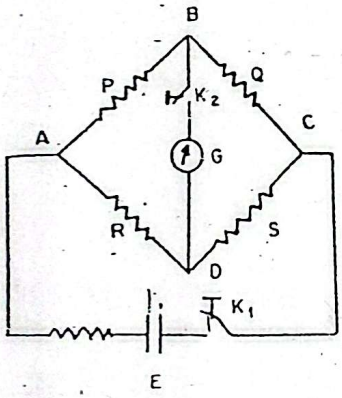

Fig. 7.30

The key K_1 is meant for battery circuit and K_2 is for the galvanometer circuit. The battery is connected to the terminal of the key K_1 and the point C. Students should note carefully whether A or C is internally connected with K_1. In some instruments C instead of A is internally connected to K_1. In that case, the battery should be connected between K_1 and the point C. The galvanometer is connected to the terminal of the key K_2 and the point D common to the arms R and S. The bridge will best be understood by comparing it with the circuital diagram (Fig. 7.30) in which the parts are lettered in the same way as in the figure of the box. The key K_1 closes the battery circuit

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